RL Reward Function: Joint-Space Skills

Biotech lab robot (Franka, Isaac Lab 3.0, PPO) · worked example: the pipette pick-up skill now in training (`rl/pipette_grasp_joint`) · 5 Nov 2026

Every new no-IK skill uses the same three-part reward. The policy commands the 7 arm joints and the gripper directly. It is paid for progress, receives a large bonus or penalty at the end of the episode, and pays small per-step costs for unsafe or jerky behaviour.

```
r = [Φ(s') − Φ(s)]      progress: paid only for improving the state
  + 100 · success        once, at the end
  −  20 · fail           once, at the end
  + small per-step costs
```

1. Progress potential Φ(s)

Φ scores how far along the pick the current state is. Each step the policy is paid the change in Φ: getting closer earns reward, backing off costs it, and standing still earns nothing. The shaping is undiscounted (`shaping_gamma = 1.0`). An earlier version used the PPO discount (0.99), which quietly charged the policy 1% of Φ for every step it waited near the target. The centrifuge-remove policy learned to swing the arm away to avoid that charge.

Notation: `d` is the distance from the hand (TCP) to the expert's grasp point on the pipette body. Most terms have the form `1 − tanh(x / scale)`: 1 at the target, falling toward 0 at a few times the scale.

Term	Weight	Measures	Why it is there
`rew_far`	3	`1 − tanh(d / 40 cm)`	A pull from starts 30–60 cm away, where the short-range terms are flat.
`rew_reach`	2	`1 − tanh(d / 10 cm)`	Coarse approach.
`rew_line`	2	`1 − tanh(lat / 2 cm)`	`lat` is the sideways distance from the approach line: a line through the grasp point along the pipette's local z, the way the expert comes in.
`rew_line_fine`	3	`1 − tanh(lat / 3 mm)`	Precise centring on that line.
`rew_down`	3	`1 − tanh(h / 1.5 cm)`, only while on the line (`lat < 5 mm`)	`h` is the height above the grasp point along the line. Descending is paid only once lined up, so the hand does not come in from the side and shove the pipette.
`rew_fine`	4	`1 − tanh(d / 1 cm)`	The final centimetre.
`rew_align`	3	`1 − tanh(rot / 0.2 rad)`, faded with distance	`rot` is the angle between the hand's orientation and the expert's grasp frame. The IK teacher set this orientation itself; the joint-space policy has to learn it.
`rew_grasp`	5	`held` (0 or 1)	Gripper closed, finger gap 15–21 mm (the 19 mm body), within 2 cm of the grasp point.

`rew_lift`	10	`held × min(lift / 3 cm, 1) × steady`	`lift` is the height the pipette has gained. `steady` falls from 1 to 0 as the pipette turns 0→20° in the hand, so a tilted lift scores little.

Φ tops out around 35, so the full approach, grasp and lift are worth about +35 in shaping.

2. Terminal rewards (once per episode)

- `rew_success = +100`: held, lifted at least 3 cm, turned less than 10°, and kept that way for **10 consecutive steps** (`success_hold_steps`, about 0.7 s). The hold requirement came from the tube pick, where a single-step success let a yanked, tilted lift count.
- `rew_fail = -20`, any of:
  - the pipette knocked over (turned more than 30° while not held);
  - it is shoved more than 5 cm without being held;
  - any neighbouring object (tray, tip rack, waste bin) moves.

  The generator fails a trial in all of these cases.

3. Per-step costs

Term	Value	Applies when
`rew_turn`	−0.1 per degree	Held and turned beyond 10°.
`rew_early_close`	−0.2	Gripper closed more than 1.5 cm from the grasp point.
`rew_near_speed`	−2 per m/s	The open hand is within 5 cm and moving faster than 0.15 m/s. This stops it sweeping into the pipette; it was added after the tube pick's knock-overs.
`rew_action_rate`	$-0.005 \times \lVert a_t - a_{t-1} \rVert^2$	Every step; discourages jerky motion.

4. Settings that shape learning (not rewards)

Setting	Value	Meaning
`joint_step_rad`	0.02	The most any joint target moves per 15 Hz step (0.3 rad/s).
`grip_threshold`	−0.8	The gripper closes when its output is above −0.8. An untrained policy is therefore mostly closed, so held lifts get explored. In tube-pick run 9 this was what made the lift learnable.
`held_p`	0.4	Reverse curriculum: 40% of episodes start with the pipette already in the hand (498 expert frames), so the lift is learned early.
Episode length	12 s	180 control steps at 15 Hz.

5. How the other skills differ

All seven skills use the same template; only the Φ terms and the failure tests change.

Skill	Φ terms and special rules
Tube pick	Hand vertical, centred, descend, grasp, lift × upright. Success: the 6 cm lift held for 1 s at under 3.5° tilt. A per-step cost when the held tube's tilt rises.
Centrifuge remove	Approach line along the tube's 45° axis, closing along the rotor's tangent, grasp, extraction progress along the bore, then lift × upright.
Centrifuge insert	Carry; on-axis (5 mm and 1 mm); coaxial; depth; in bore; released seated; backed off. Pays tube speed down the bore axis in the entry zone. Release interlock: the gripper may open only with the tube inside the bore.
Tray pick	Hand pointing down, closing axis across the tray, held, lift × level.
Tray place	Carry, yaw, level, lower, released, backed off. Release interlock: opening only within 1 cm of the bench at the target spot.
Tip attach, pipetting position	On the axis (5 mm and 0.5 mm scales), depth while on the axis, alignment. No gripper action; the pipette stays held.

> Source: `rl/pipette_grasp_joint/env.py` (reward at line 72) and `env_cfg.py`; the shared joint-space action layer is `rl/joint_space.py`. Inherited thresholds (hold gap, lift goal, failure limits) come from `rl/pipette_grasp/env_cfg.py`.